Concordia University · Gina Cody School of Engineering · Department of Mechanical, Industrial and Aerospace Engineering

INDU 211: Production & Manufacturing Systems

Midterm Exam Master Study Guide & Complete Solved Examination (Paper A)

Fall 2025 Session 75 Minutes (Closed Book) Total Marks: 50 Chapters 1–5 Mastery
Pedagogical Master Architecture: This compendium synthesizes teacher lecture notes, Callister & Groover curriculum topics, and the authentic Fall 2025 Midterm Exam Paper A. Every question features step-by-step mathematical proofs, official grading keys, distractor eliminations, and detailed diagnostic post-mortems of common exam failure modes.

1. Executive Blueprint & Diagnostic Scorecard

Problem Module	Curriculum Scope	Format & Tasks	Marks	Weight
Problem 1	Chapters 1, 2, 3, 4, 5	12 Concept & Quantitative MCQs (1 mark each)	12 M	24%
Problem 2	Chapter 3 (Manufacturing)	Multi-Level Bill of Materials (BOM) Explosion Table	10 M	20%
Problem 3	Chapter 4 (Facility Logistics)	Transportation LP (Northwest Corner, Lowest Cost, Optimality)	16 M	32%
Problem 4	Chapter 5 (Material Handling)	AGV Fleet Routing (Asymmetric TSP Nearest-Neighbor)	12 M	24%
Total Examination Value			**50 M**	**100%**

2. Problem 1: Core Multiple Choice Solutions (12 Marks)

Question 1: Nature of Science vs. Engineering Practice Correct Answer: (c)

Statement: Which of the following statements is correct?

- *(a) Science is based purely on imagination and creativity rather than observation and experimentation.* [False: Science is empirically rooted in the scientific method].
- *(b) Results generated from computer modeling are sufficiently realistic to guide engineering practice.* [False: Computer models are idealized approximations requiring physical validation].
- **(c) Scientific theories can be validated through physical experiments, and these theories may be applied to develop technologies that solve practical problems.** [CORRECT].

Core Principle: Science discovers fundamental physical laws through empirical experimentation; engineering synthesizes these principles into safe, cost-effective technologies.

Question 2: Statutory Scope of Engineering Practice Correct Answer: (d)

Statement: Which of the following activities falls under the "practice of engineering"?

- (a) Designing a bridge using structural analysis principles.
- (b) Advising on safe material use for a factory process.
- (c) Supervising the construction of a power plant.
- **(d) All of the above.** [CORRECT: Statutory provincial law encompasses design, advisory, and technical supervision affecting public safety].

Question 3: Professional Engineering Regulation in Canada Correct Answer: (b)

Statement: In Canada, engineering is a profession and:

- *(a) Can be carried out by scientists who hold a Ph.D. in a scientific field.* [False: A doctorate does not grant license to practice].
- **(b) Is regulated by provincial associations of professional engineers.** [CORRECT: Provincial jurisdiction via 12 constituent bodies, e.g., OIQ in Quebec, PEO in Ontario].
- *(c) Is open to anyone with a high school diploma and practical skills.* [False: Closed profession requiring accredited CEAB degree].
- *(d) Is governed by UNCSTD.* [False: UNCSTD is an international advisory body, not a licensing regulator].

Question 4: Scope of Engineering vs. Pure Science (Historic Trap) Correct Answer: (a)

Statement: Which of the following is NOT necessarily part of the engineering profession as defined in its practice?

- **(a) Performing scientific experiments to establish new theories.** [CORRECT ANSWER: This is the definitive mandate of pure science, not engineering practice].
- *(b) Safeguarding life and health.* [Mandatory legal obligation under the code of ethics].
- *(c) Protecting economic interests and property.* [Core duty: designing economically viable and safe systems; classic student trap].
- *(d) Ensuring public welfare and environmental safety.* [Paramount duty of all professional engineers].

Exam Trap Post-Mortem: Students frequently select (c) believing engineers ignore economics. In reality, economic viability is an explicit duty, whereas establishing fundamental scientific theories belongs exclusively to pure science.

Question 5: IISE Definition of Industrial Engineering Correct Answer: (c)

Statement: Industrial Engineering, as a discipline, mainly deals with issues related to:

- **(c) Analyzing, designing, and evaluating integrated systems of people, equipment and information.** [CORRECT: Official IISE definition of integrated systems of people, materials, equipment, energy, and information].
- *(a) Social app development* (Software Eng); *(b) Chemical synthesis* (Biochem/Pharm); *(d) Microchip fabrication* (Electrical Eng).

Question 6: Classification of Human Activity Systems Correct Answer: (b)

Statement: Which of the following is an example of a human activity system considered by IE?

- **(b) Physical workplace layout, material handling, safety procedures.** [CORRECT: Ergonomic, anatomical, and physical interaction between workers and physical facilities].
- *(a) Forecasting* (Mathematical model); *(c) Scheduling* (Planning/control); *(d) Budgeting* (Managerial financial system).

Question 7: Single-Product Break-Even Pricing Calculation Correct Answer: (d)

Problem: Fixed Cost $FC = \$15,000$, Variable Cost $VC = \$30/\text{{unit}}$, Break-even Quantity $Q_{{BE}} = 500\text{{ units}}$. Find unit selling price P for profitability:

Q_{{BE}} = \frac{{FC}}{{P - VC}} \implies P - VC = \frac{{FC}}{{Q_{{BE}}}} \implies P = VC + \frac{{FC}}{{Q_{{BE}}}} = 30 + \frac{{15,000}}{{500}} = 30 + 30 = \mathbf{{\$60/\text{{unit}}}}

Validation: Total Revenue $TR = 60 \times 500 = \$30,000$. Total Cost $TC = 15,000 + (30 \times 500) = \$30,000$. Profit $\pi = TR - TC = \$0$. Any price $P \ge \$60$ ensures profitability.

Question 8: Multi-Process Selection via Crossover Volume Correct Answer: (a)

Problem: Projected annual demand is $Q = 5,200\text{{ units}}$. Select between:

- **Process A:** $FC = \$50,000$, $VC = \$10/\text{{unit}}$
- **Process B:** $FC = \$80,000$, $VC = \$5/\text{{unit}}$

\text{{Crossover Volume: }} Q^* = \frac{{\Delta FC}}{{\Delta VC}} = \frac{{80,000 - 50,000}}{{10 - 5}} = \frac{{30,000}}{{5}} = \mathbf{{6,000\text{{ units}}}}

Evaluation at $Q = 5,200 < 6,000$:

- $TC_A = 50,000 + 10(5,200) = \mathbf{{\$102,000}}$
- $TC_B = 80,000 + 5(5,200) = \mathbf{{\$106,000}}$
- **Decision:** Process A is preferred because it yields a net savings of $\$4,000$ (reaches break-even earlier). Selecting B purely for lower variable cost is a catastrophic trap.

Question 9: Key Advantage of Concurrent Engineering Correct Answer: (c)

Answer: (c) Faster time to market and better product quality. Cross-functional collaboration replaces sequential "over-the-wall" engineering, minimizing late change orders and shortening product launch lead time.

Question 10: Three Basic Types of Facility Layout Correct Answer: (b)

Answer: (b) Product layout, process layout, fixed layout. Product layout (flow line), process layout (functional job shop), and fixed-position layout (stationary product project).

Question 11: Skill vs. Machine Investment Trade-off Correct Answer: (b)

Answer: (b) Product layout. Highly automated, specialized machinery (high capital investment) performs short-cycle repetitive operations requiring minimal worker craft skill.

Question 12: Material Handling Modernization Category Correct Answer: (a)

Answer: (a) Changes in methods and equipment. Introducing automated guided vehicle (AGV) systems is a technological modernization of material handling methods and equipment.

3. Problem 2: Multi-Level Bill of Materials (BOM) Explosion (10 Marks)

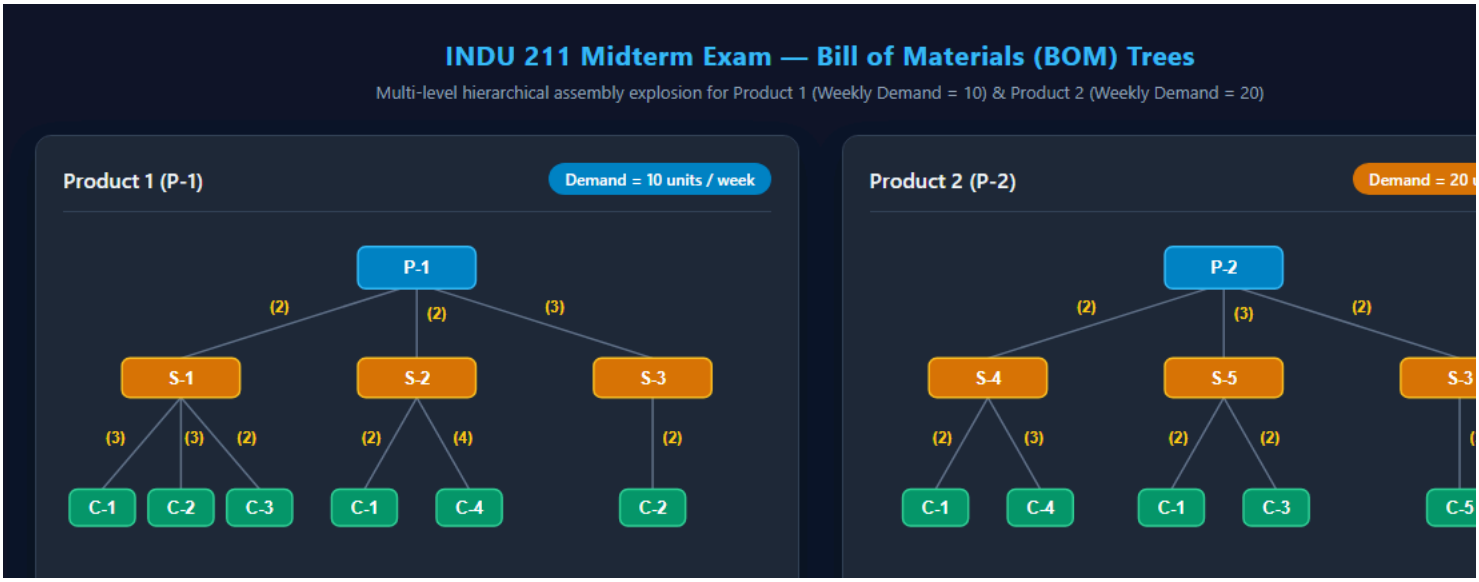


Figure 1: Multi-level Bill of Materials hierarchy for Product 1 (Weekly Demand = 10) and Product 2 (Weekly Demand = 20).

Step-by-Step Mathematical Explosion

Weekly Demands: $D(P_1) = 10\text{{ units/week}}$, $D(P_2) = 20\text{{ units/week}}$.

1. Intermediate Subassembly Requirements:

- **S-1:** Used only in P_1 with multiplier 2 $\implies 10 \times 2 = \mathbf{{20\text{{ units}}}}$
- **S-2:** Used only in P_1 with multiplier 2 $\implies 10 \times 2 = \mathbf{{20\text{{ units}}}}$
- **S-3 (Common Subassembly):**
  - From P_1: $10 \times 3 = 30\text{{ units}}$
  - From P_2: $20 \times 2 = 40\text{{ units}}$
  - Total $S_3 = 30 + 40 = \mathbf{{70\text{{ units}}}}$
- **S-4:** Used only in P_2 with multiplier 2 $\implies 20 \times 2 = \mathbf{{40\text{{ units}}}}$
- **S-5:** Used only in P_2 with multiplier 3 $\implies 20 \times 3 = \mathbf{{60\text{{ units}}}}$

2. Raw Component Downward Explosion:

- **C-1:** Required across $S_1(3), S_2(2), S_4(2), S_5(2)$: $$\text{{Total }} C_1 = (20 \times 3) + (20 \times 2) + (40 \times 2) + (60 \times 2) = 60 + 40 + 80 + 120 = \mathbf{{300\text{{ units}}}}$$
- **C-2:** Required in $S_1(3)$ and S_3 under $P_1(2)$: $$\text{{Total }} C_2 = (20 \times 3) + (30 \times 2) = 60 + 60 = \mathbf{{120\text{{ units}}}}$$
- **C-3:** Required in $S_1(2)$ and $S_5(2)$: $$\text{{Total }} C_3 = (20 \times 2) + (60 \times 2) = 40 + 120 = \mathbf{{160\text{{ units}}}}$$
- **C-4:** Required in $S_2(4)$ and $S_4(3)$: $$\text{{Total }} C_4 = (20 \times 4) + (40 \times 3) = 80 + 120 = \mathbf{{200\text{{ units}}}}$$
- **C-5:** Required in S_3 under $P_2(3)$: $$\text{{Total }} C_5 = 40 \times 3 = \mathbf{{120\text{{ units}}}}$$

Official Master Solution Table (1 Mark per Correct Cell = 10 Marks)

P-1	P-2	S-1	S-2	S-3	S-4	S-5	C-1	C-2	C-3	C-4	C-5
10	20	20	20	70	40	60	300	120	160	200	120

4. Problem 3: Transportation Linear Programming & Heuristics (16 Marks)

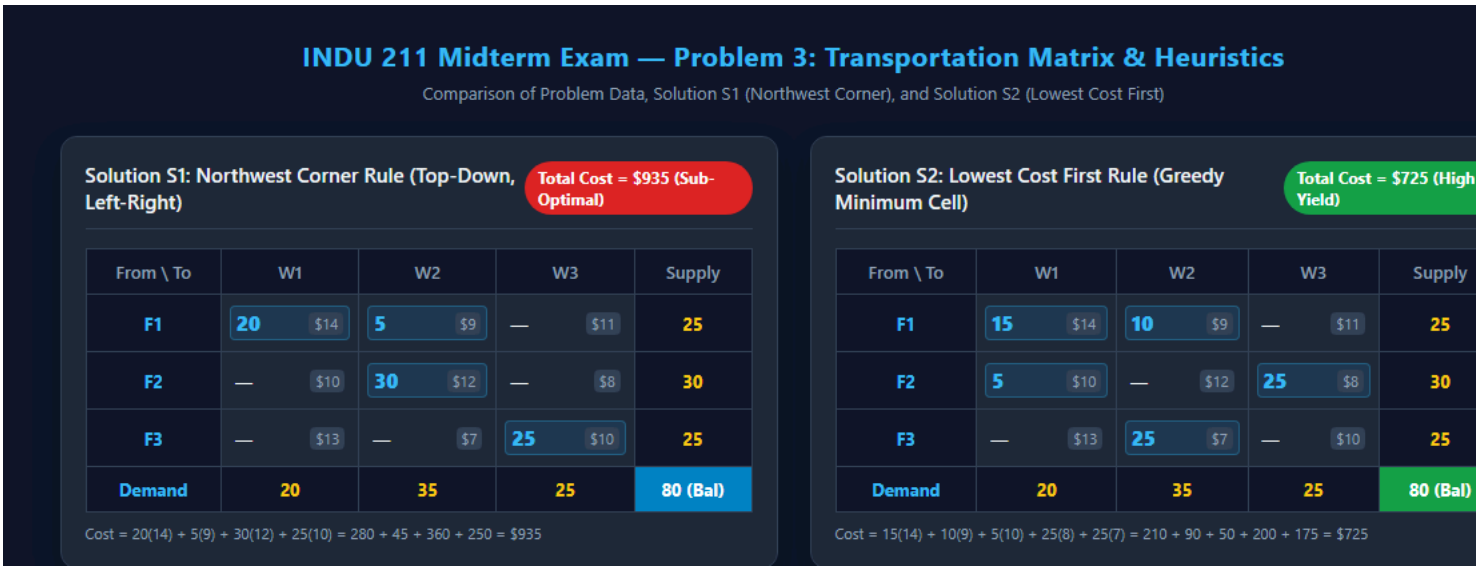


Figure 2: Transportation initial matrix and side-by-side comparison of Solution S1 (Northwest Corner, Cost = \$935) vs. Solution S2 (Lowest Cost First, Cost = \$725).

Balanced System Verification

$$\sum_{{i=1}}^3 a_i = 25 + 30 + 25 = 80 \quad \text{{and}} \quad \sum_{{j=1}}^3 b_j = 20 + 35 + 25 = 80 \implies \mathbf{{Balanced\ System}}$$

Solution S1: Northwest Corner Rule (Top-Down, Left-Right)

1. Cell (F_1, W_1): Allocate $\min(25, 20) = 20$. Warehouse W_1 satisfied. F_1 has 5 left.
2. Cell (F_1, W_2): Allocate $\min(5, 35) = 5$. Factory F_1 exhausted. W_2 has 30 remaining.
3. Cell (F_2, W_2): Allocate $\min(30, 30) = 30$. Both F_2 and W_2 satisfied simultaneously.
4. Cell (F_3, W_3): Allocate $\min(25, 25) = 25$. All constraints fully met.

\mathbf{{3.1\ Total\ Transportation\ Cost\ of\ S1}} = 20(14) + 5(9) + 30(12) + 25(10) = 280 + 45 + 360 + 250 = \mathbf{{\$935}}

Solution S2: Lowest Cost First Rule (Greedy Minimum Cell)

1. Min cost is $c_{{32}} = 7$: Allocate $\min(25, 35) = 25$ to (F_3, W_2). Factory F_3 exhausted. W_2 needs 10.
2. Next min cost is $c_{{23}} = 8$: Allocate $\min(30, 25) = 25$ to (F_2, W_3). Warehouse W_3 satisfied. F_2 has 5 left.
3. Next min cost is $c_{{12}} = 9$: Allocate $\min(25, 10) = 10$ to (F_1, W_2). Warehouse W_2 satisfied. F_1 has 15 left.
4. In remaining Column W_1, lowest cost is $c_{{21}} = 10$: Allocate $\min(5, 20) = 5$ to (F_2, W_1). F_2 exhausted. W_1 needs 15.
5. Final allocation to cell (F_1, W_1) with cost 14: Allocate $\min(15, 15) = 15$.

\mathbf{{3.3\ Total\ Transportation\ Cost\ of\ S2}} = 15(14) + 10(9) + 5(10) + 25(8) + 25(7) = 210 + 90 + 50 + 200 + 175 = \mathbf{{\$725}}

Question 3.4: Optimality Analysis (True, False, or MTF)

- **A. S1 is an optimal solution of this transportation problem:** F (False)
  Proof: Feasible solution S2 achieves $\$725 < \$935$. By definition, an optimal solution minimizes cost over all feasible solutions. Since a strictly cheaper feasible solution exists, S1 cannot be optimal.
- **B. S1 is NOT an optimal solution of this transportation problem:** T (True)
  Proof: Logical negation of Statement A. Since S1 is sub-optimal, stating that it is NOT optimal is an established mathematical truth.
- **C. S2 is an optimal solution of this transportation problem:** MTF (May be True or False)
  Proof: Lowest Cost First is a greedy heuristic that provides a starting basic solution, not a guaranteed global optimum. Without executing the MODI ($u-v$) method, one cannot verify whether S2 is optimal. (Indeed, MODI reveals a true global optimum of $\$710$, proving S2 is technically sub-optimal).
- **D. This problem is a balanced transportation problem:** T (True)
  Proof: Total supply ($\sum a_i = 80$) exactly balances total demand ($\sum b_j = 80$).

5. Problem 4: Traveling Salesperson Problem (TSP) & AGV Routing (12 Marks)

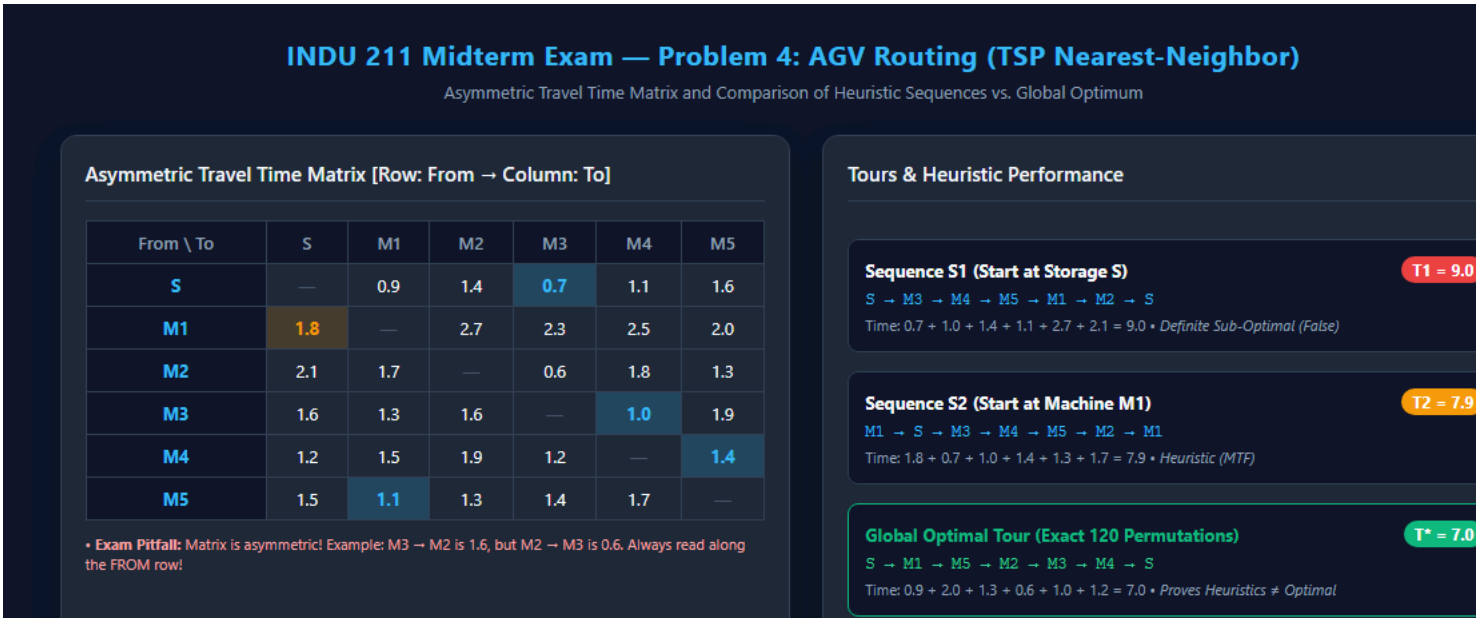


Figure 3: AGV travel time matrix, directional asymmetry warning, and routing sequence performance comparison.

Sequence S1: Nearest-Neighbor from Storage Room S

1. Start at S: Destinations $\{M_1..M_5\}$. Row S min is M_3 with time **0.7**.
2. At M_3: Available $\{M_1, M_2, M_4, M_5\}$. Row M_3 min is M_4 with time **1.0**.
3. At M_4: Available $\{M_1, M_2, M_5\}$. Row M_4 min is M_5 with time **1.4**.
4. At M_5: Available $\{M_1, M_2\}$. Row M_5 min is M_1 with time **1.1**.
5. At M_1: Only M_2 remains unvisited. Row M_1 time to M_2 is **2.7**.
6. Return from M_2 to S: Row M_2 time to S is **2.1**.

\mathbf{{S1}} = S \rightarrow M_3 \rightarrow M_4 \rightarrow M_5 \rightarrow M_1 \rightarrow M_2 \rightarrow S \qquad \mathbf{{T1}} = 0.7 + 1.0 + 1.4 + 1.1 + 2.7 + 2.1 = \mathbf{{9.0\text{{ minutes}}}}

Sequence S2: Nearest-Neighbor from Machine Center M1

1. Start at M_1: Destinations $\{S, M_2..M_5\}$. Row M_1 min is S with time **1.8**.
2. At S: Destinations $\{M_2..M_5\}$. Row S min is M_3 with time **0.7**.
3. At M_3: Destinations $\{M_2, M_4, M_5\}$. Row M_3 min is M_4 with time **1.0**.
4. At M_4: Destinations $\{M_2, M_5\}$. Row M_4 min is M_5 with time **1.4**.
5. At M_5: Only M_2 remains. Row M_5 time to M_2 is **1.3**.
6. Return from M_2 to origin M_1: Row M_2 time to M_1 is **1.7**.

\mathbf{{S2}} = M_1 \rightarrow S \rightarrow M_3 \rightarrow M_4 \rightarrow M_5 \rightarrow M_2 \rightarrow M_1 \qquad \mathbf{{T2}} = 1.8 + 0.7 + 1.0 + 1.4 + 1.3 + 1.7 = \mathbf{{7.9\text{{ minutes}}}}

Question 4.3: Optimality Evaluation

- **a) S1 is optimal:** F (False) — Sequence S2 completes all deliveries in $7.9 < 9.0\text{{ min}}$. S1 is strictly sub-optimal.
- **b) S2 is optimal:** MTF (May be True or False) — Nearest-neighbor is a greedy heuristic that does not guarantee global optimality.
- **c) S1 is NOT optimal:** T (True) — Direct complement of (a); confirmed by existence of faster tour S2.
- **d) S2 is NOT optimal:** MTF (May be True or False) — Without evaluating all 120 tours, asserting it is definitely not optimal is also unverified.

Question 4.4: Multi-Start Heuristic Principle

Master Principle: TSP is NP-hard. Generating multi-start heuristic tours from all N nodes improves solution quality, but no greedy heuristic is guaranteed to find the true global minimum over all $(N-1)!$ combinatorial tours. Exhaustive calculation reveals the true global minimum tour is $S \rightarrow M_1 \rightarrow M_5 \rightarrow M_2 \rightarrow M_3 \rightarrow M_4 \rightarrow S$ with time $T^* = \mathbf{{7.0\text{{ minutes}}}}$.